

LLaMa Architecture

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Intelligent Systems
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Outline

1. Transformer
2. GPT-3
3. LLaMa
4. Pre-normalization of Input
5. SwiGLU Activation
6. Rotary Positional Embeddings (RoPE)
7. How we can improve?

Transformer

1. Input:

$$\mathbf{x} \in \mathbb{R}^{n \times d_{\text{model}}}, \quad \text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$$

$$\mathbf{h}^{(0)} = \mathbf{E} + \text{PE} \in \mathbb{R}^{n \times d_{\text{model}}}$$

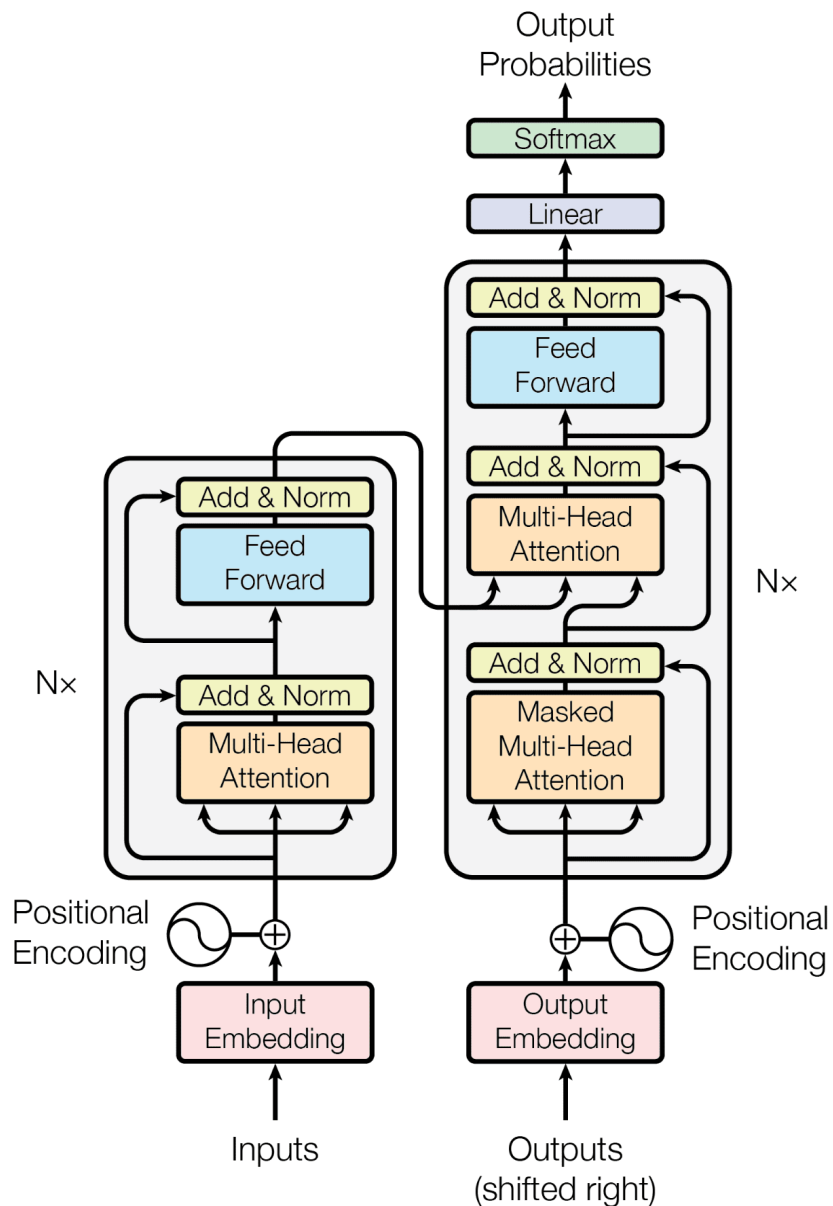
2. Multi-Head Attention & Norm:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}$$

$$\text{head}_i = \text{Attention}(\mathbf{h}^{(0)} \mathbf{W}_i^Q, \mathbf{h}^{(0)} \mathbf{W}_i^K, \mathbf{h}^{(0)} \mathbf{W}_i^V)$$

$$\mathbf{Z} = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O, \quad \mathbf{W}^O \in \mathbb{R}^{h \cdot d_v \times d_{\text{model}}}$$

$$\mathbf{h}^{(1)} = \text{LayerNorm}(\mathbf{h}^{(0)} + \mathbf{Z})$$



¹Vaswani, A., et al (2017). Attention Is All You Need.

GPT-3

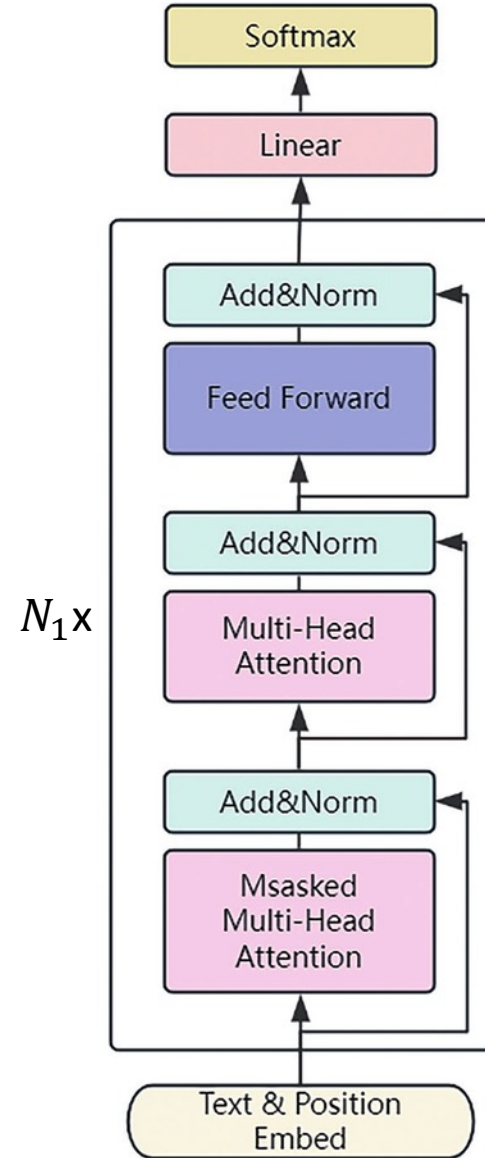
1. Autoregressive nature
2. No encoder
3. Increased complexity

Transformer:

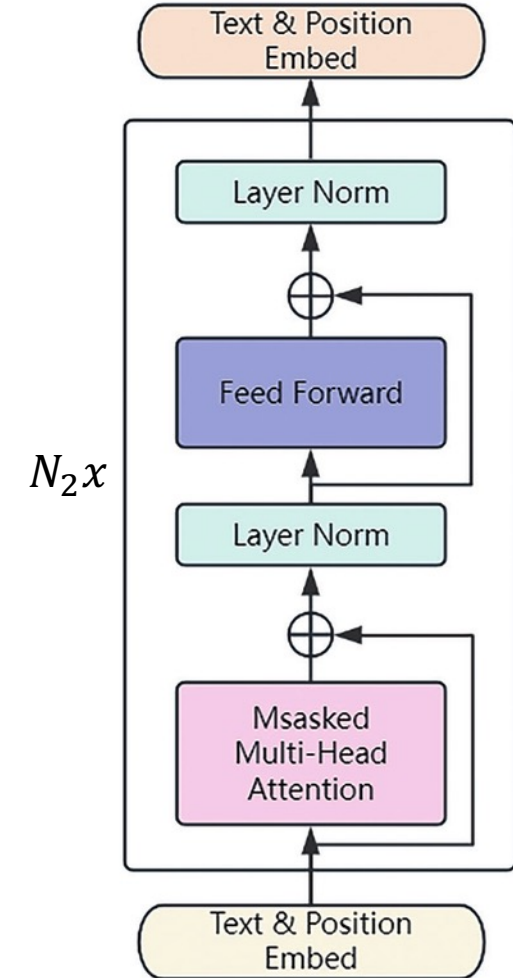
$$d_{model_1} = 1\,024 \quad N_1 = 6$$

GPT-3:

$$d_{model_2} = 12\,228 \quad N_2 = 96$$



(a) Original Transformer Decoder



(b) GPT Transformer Decoder (GPT-3)

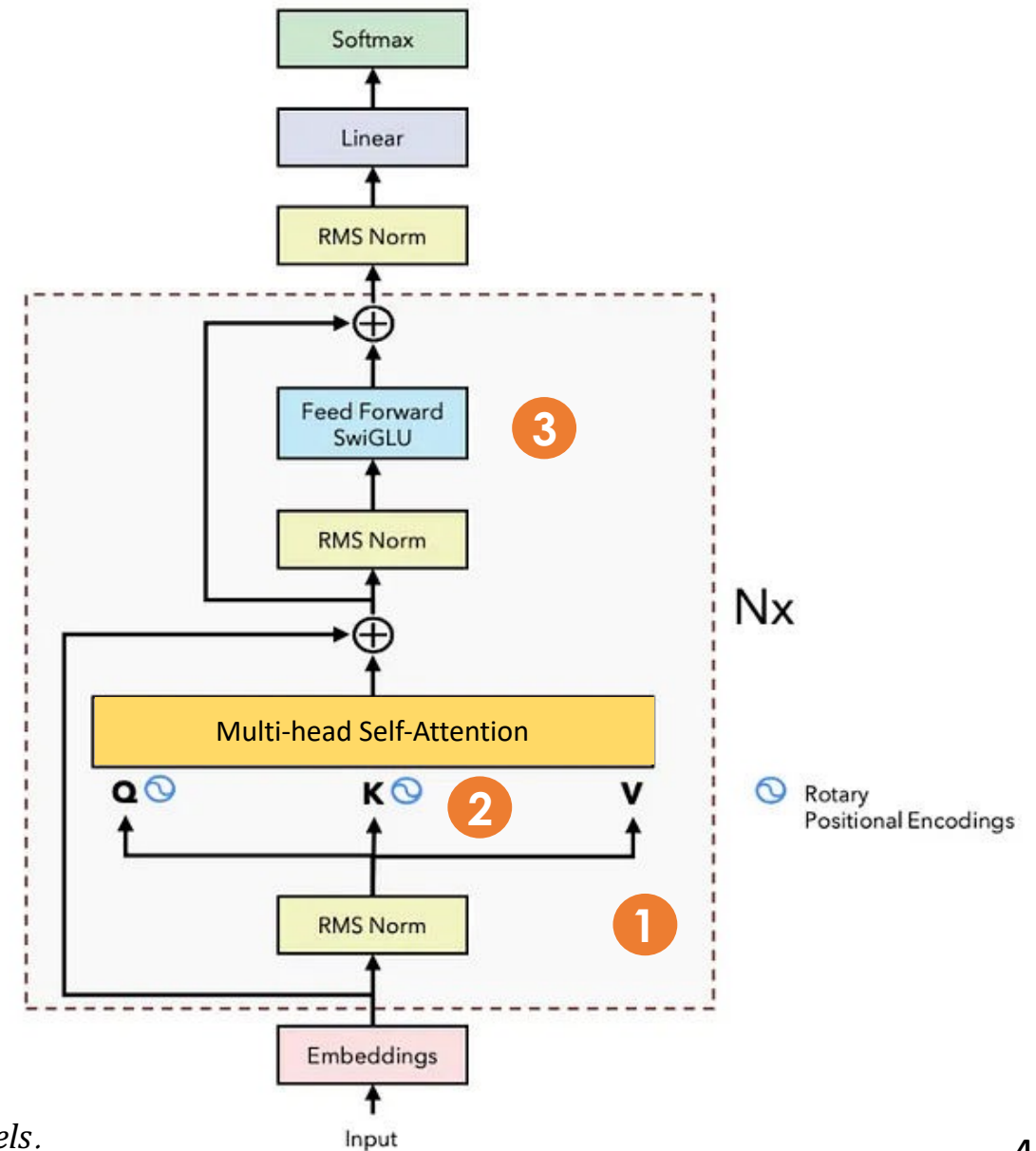
²Brown, T., et al (2020). Language Models are Few-Shot Learners.

LLaMa Architecture

LLaMa (Large Language Model Meta AI)

Available: 7B, 13B, 32B, 64B

- 1. Pre-normalization of Input
- 2. Rotary Positional Embeddings
- 3. SwiGLU



³Touvron, H., et al (2023). LLaMA: Open and Efficient Foundation Language Models.

Pre-normalization of Input

Given an $\mathbf{x} \in \mathbb{R}^m$, FFN projects it into $\mathbf{y} \in \mathbb{R}^n$:

$$a_i = \sum_{j=1}^m w_{ij} x_j, \quad y_i = f(a_i + b_i),$$

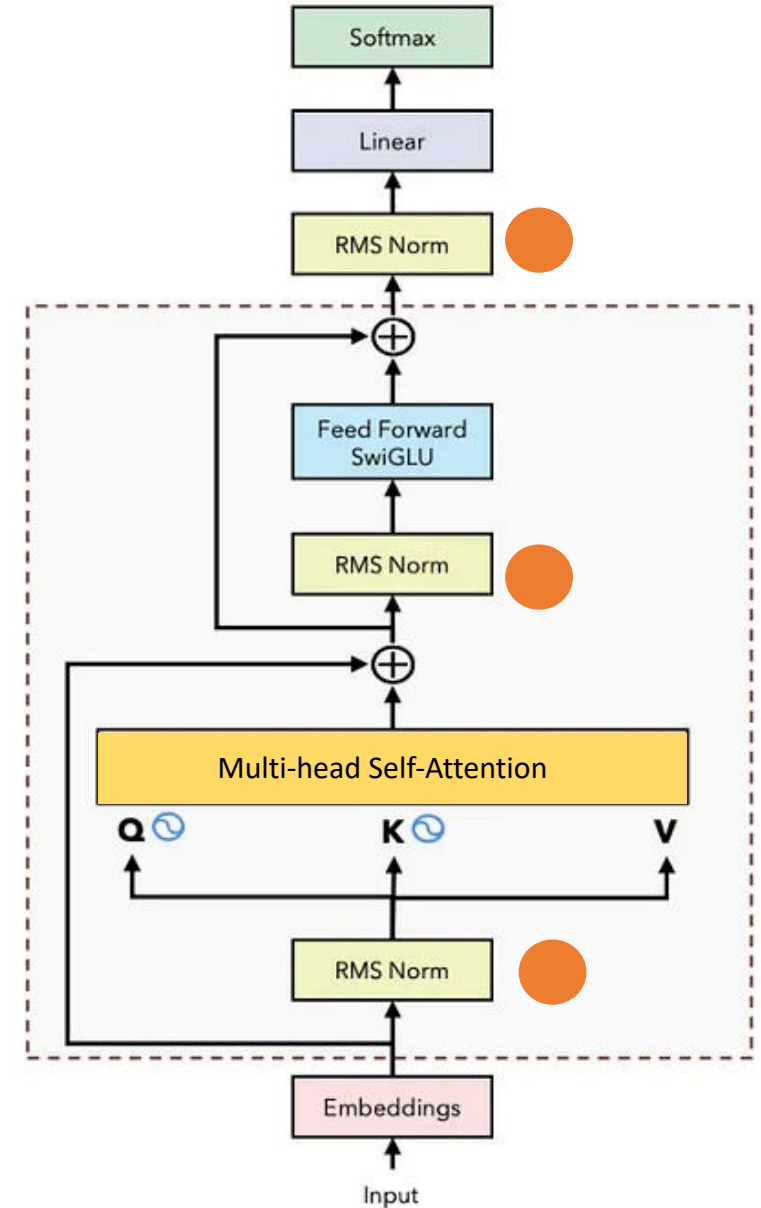
LayerNorm:

$$\bar{a}_i = \frac{a_i - \mu}{\sigma} g_i, \quad y_i = f(\bar{a}_i + b_i),$$

$$\mu = \frac{1}{n} \sum_{i=1}^n a_i, \quad \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - \mu)^2}.$$

RMSNorm:

$$\bar{a}_i = \frac{a_i}{\text{RMS}(\mathbf{a})} g_i, \quad \text{where } \text{RMS}(\mathbf{a}) = \sqrt{\frac{1}{n} \sum_{i=1}^n a_i^2}.$$



⁴Zhang, B., et al (2019). RMSNorm: Root Mean Square Layer Normalization.

Rotary Positional Embeddings

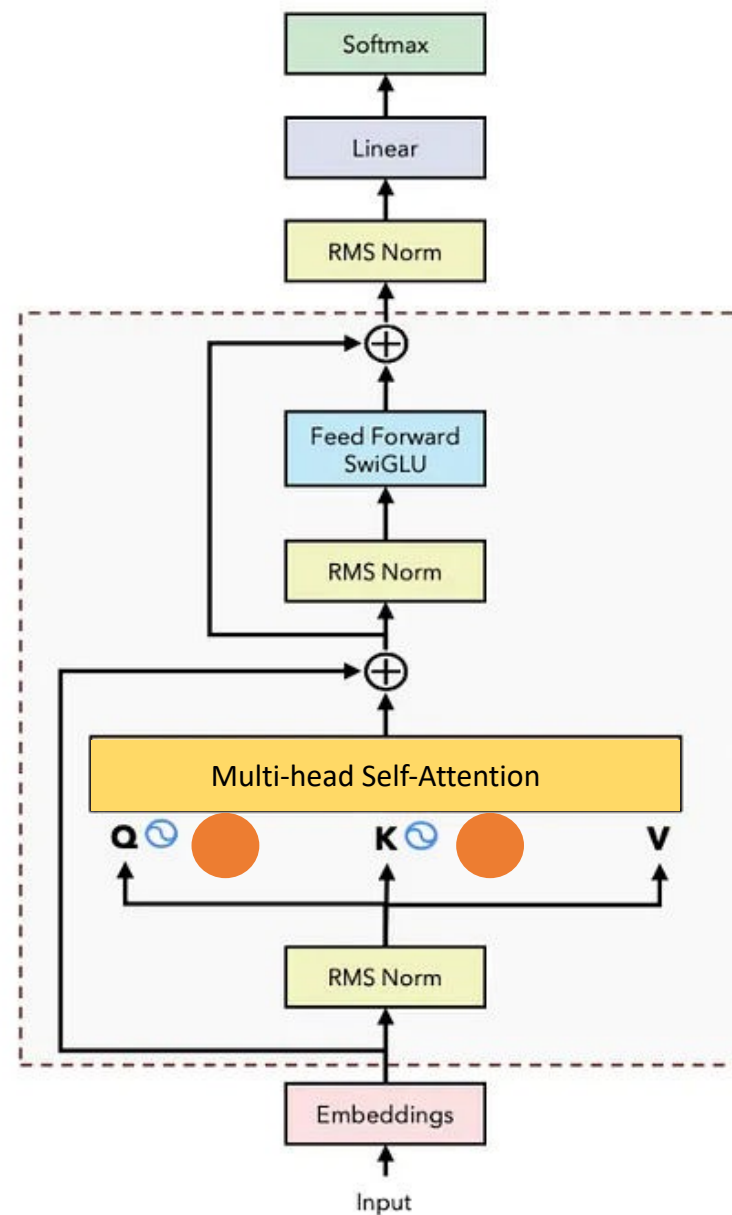
Sinusoidal Positional Encoding:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

Rotary Positional Embedding (RoPE):

$$\text{RoPE}(x_{pos}) = x_{pos} \cdot \cos(\theta_{pos}) + \hat{x}_{pos} \cdot \sin(\theta_{pos})$$



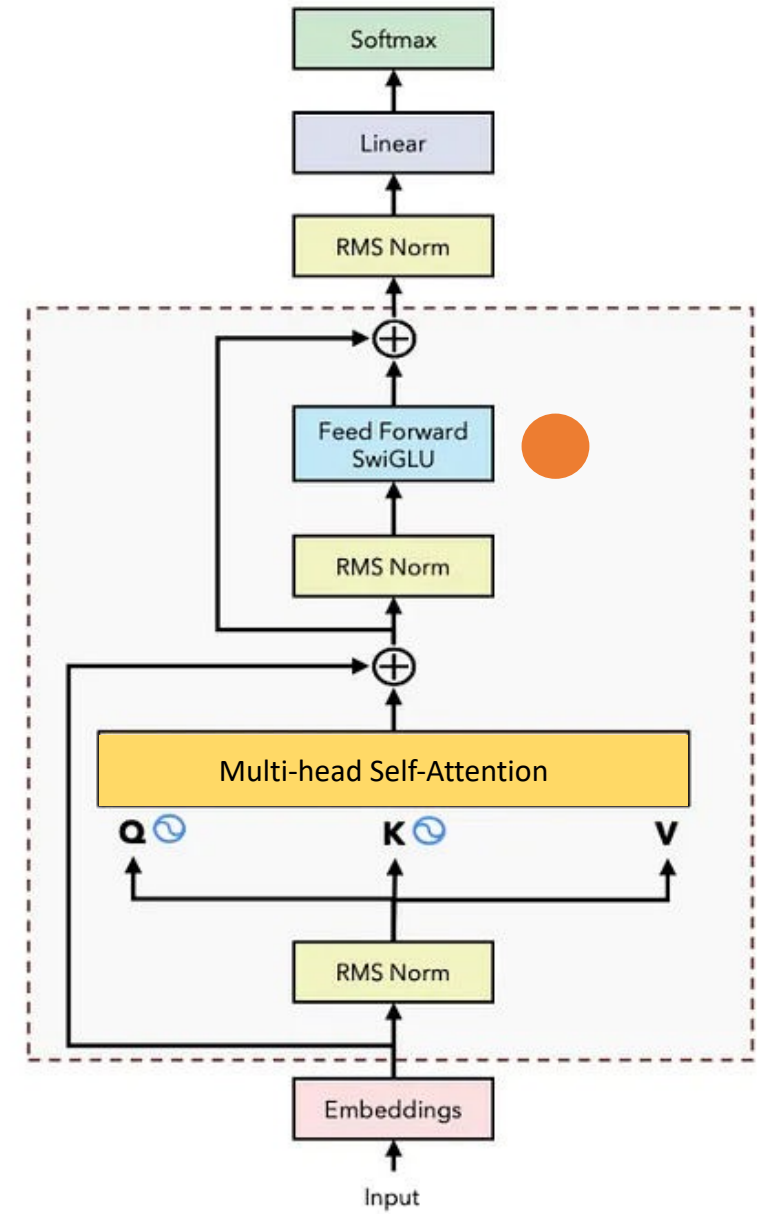
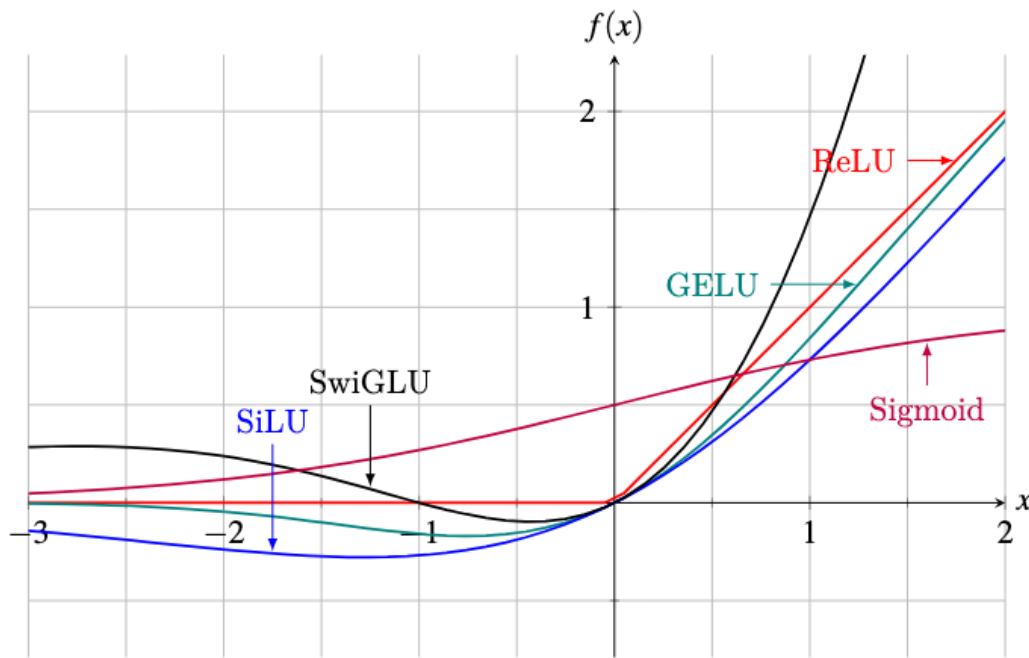
⁵Su, J., et al (2021). RoFormer: Enhanced Transformer with Rotary Position Embedding.

SwiGLU

$$\text{ReLU}(x) = \max(0, x)$$

$$\text{GLU}(x) = x * \sigma(Wx + b)$$

$$\text{SwiGLU}(x) = x\sigma(\beta x) + (1 - \sigma(\beta * x)) * (Wx + b)$$



⁶Shazeer, N. (2020). GLU Variants Improve Transformer.

What we can improve?

1. Multi-head Attention / Grouped-query Attention / Multi-query
2. FlashAttention
3. KV Cache
4. Scaling Law
5. ...